

VAIBHAV JAIN

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EDUCATION

BITS Pilani, Hyderabad Campus

Expected July 2027

B.E. Electrical & Electronics Engineering

SKILLS & TOOLS

Languages: Java, Python, SQL

Backend & Infra: Spring Boot, PostgreSQL, Redis, Apache Kafka, AWS, Docker, CI/CD (GitHub Actions)

AI Engineering: Gemini API (text + vision), LangChain, RAG pipelines, multi-modal LLM evaluation, prompt engineering, schema-validated JSON, context caching; AI-assisted development with Claude Code / Cursor

Data & ML: Pandas, DuckDB, LightGBM, SHAP, NetworkX

CERTIFICATIONS

Cloud Computing Foundations — Duke University (Coursera, 2026) · Nextleap PM Fellowship (Cohort 45)

WORK EXPERIENCE

Wiingy · AI Engineering Intern · GenAI Systems | EdTech platform | AI-assisted tutor hiring

Apr 2026 – Present

- Built and deployed a **multi-modal AI tutor evaluation service** (FastAPI on AWS EC2, consumed by Wiingy's main backend) that **auto-rejects ~60% of incoming applicants in ~2–3 min per submission** with structured PASS / FLAG / REJECT verdicts and per-modality reasoning across text, 5 images, audio, and video — collapsing a manual hiring-review queue previously handled by a 4-person ops team.
- Engineered the inference layer over **Gemini 2.5 Flash Lite** with TTL-based **context caching** of system prompts, deterministic config (T=0, seed=42), explicit **thinking-budget allocation (2,048 tokens)** for safety-critical evaluations, **ThreadPoolExecutor parallel fan-out** across modalities, and exponential-backoff retry with json-repair fallback for malformed output — deterministic parse failures intentionally short-circuit the retry loop to avoid wasted tokens.
- Designed **6 specialized prompt evaluators** across 4 modalities with ~50+ evaluation criteria (cascading rules, cultural-inclusivity guardrails) and **two-layer output validation** (schema-enforced JSON + secondary LLM verifier); calibrated via **iterative prompt refinement on a 1,000-tutor / 6,000+ image dataset** with override-feedback loops across multiple versions per modality, and shipped a **two-tier aggregation** (per-input → per-tutor) with thresholds re-tunable on saved verdicts without LLM re-invocation.

The Mirror · Founder's Office Intern | Early-stage consumer tech startup

Aug – Nov 2025

- Analysed notification engagement across **3 alert formats** and shipped LLM-powered personalisation that **doubled click-to-open rate**; debugged a notification-truncation bug capping top-of-funnel acquisition.
- Instrumented an onboarding funnel in SQL across a **500+ user cohort**, isolated profile creation as the primary drop-off, and drove a flow redesign that cut signup abandonment **18%**.
- Automated a weekly growth KPI pipeline across **5 core metrics**, replacing manual compilation and cutting founder review prep from **~90 → ~20 min** per week.

PROJECTS

Provenance · AI-Native Dark-Store Intake Quality Gate · Java · Spring Boot · PostgreSQL · Redis · Kafka · AWS

2026

- Built and deployed a **production-ready AI-native quality gate** for curated-grocery dark-store intake — workers submit SKU + 3 photos via POST / intake, the service returns 202 in **<100ms p95** while Gemini 2.5 Vision runs async multi-dimensional quality checks across **5 product categories × 5 quality dimensions** and writes an ACCEPT / HOLD / REJECT decision to Postgres with photographic evidence and category-specific rationale, with full decisions arriving in **~5s median**.
- Designed a **three-path Spring Boot 3 architecture** — sync intake, async Kafka-driven verification, read-side audit — over PostgreSQL + Redis on AWS; Redis caches per-SKU reference data with prompt-version-keyed invalidation, LLM timeouts default to HOLD with a human-review flag (not auto-accept) preserving the quality moat under degraded service.
- Sustained **100 intakes/sec at <100ms p95** on the metadata path under k6 load test (single t3.small EC2); full stack ships via docker-compose with **GitHub Actions CI/CD** (lint → test → build → push to ECR) and a live React dashboard streaming real-time intake feed, SKU drill-downs, and audit queries filterable by vendor / store / date / decision.

ForensIQ · AI Credit Intelligence Engine · Python · Gemini 2.5 Vision · LightGBM · SHAP · NetworkX · [\[Live →\]](#) Hackathon · 2026

- Built and deployed a **4-stage credit appraisal service** for Indian MSME lending — ingests bank statements, GST returns, and legal notices through a zero-OCR Gemini 2.5 Vision pipeline (9-class transaction taxonomy, Indian number-format parsing, retry-with-backoff against free-tier rate limits) and emits a regulator-grade Credit Appraisal Memo, compressing credit-officer turnaround from **~2 weeks → ~3 min** per application.
- Engineered the forensic core as a **deterministic + LLM hybrid** — 5 pure-Python tests (balance integrity within ₹10, GST-bank reconciliation in ±15%, NetworkX cycle detection for circular trading, transaction clustering, legal severity arithmetic) running ahead of a **4-dimension parallel research layer** (ThreadPoolExecutor + Tavily across promoter, litigation, sector, and banking signals) synthesised by Groq Llama 3.3 70B into a cross-compounding risk verdict.
- Trained a **LightGBM default-prediction model (AUC 0.996)** with **SHAP per-feature attribution** over a 5C scoring framework — every sub-score traceable to a source document and a specific test, with a hard IBC/NCLT/SARFAESI regulatory override enforcing RBI prudential norms; shipped as a self-contained Streamlit app with HTML CAM export.

ShelfWatch · Public Catalog Intelligence for Curated Grocery · Python · Playwright · DuckDB · Streamlit

2026

- Built and deployed a **scheduled catalog intelligence pipeline** that scrapes a live Indian quick-commerce catalog every 6 hours via GitHub Actions, accumulating **~10,000+ SKU snapshots over 14 days** into DuckDB and surfacing daily catalog velocity, price volatility, and stock-out frequency in a public Streamlit dashboard.
- Engineered **6 derived metrics** (per-SKU price volatility, category-level out-of-stock frequency, weekly catalog churn rate, category mix drift, catalog velocity, top weekly movers) — used the dataset to author a published operator analysis identifying three structural patterns in catalog evolution invisible from a single point-in-time view.

POSITIONS OF RESPONSIBILITY

President, Ethos — Professional Development Club, BITS Pilani Hyderabad

Aug 2025 – Present

- Lead a **30-member team** across **three concurrent initiatives** serving **500+ students**.

Member, Fest Appraisal Committee, BITS Pilani Hyderabad

2025 – 2026

- Evaluated budgets across **8+ departments** for a **₹2+ crore festival**; recommendations adopted in final execution.